

Reza Shahsavani, M. Eng., M. Sc., EIT

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Calgary, AB, Canada

SUMMARY

- M. Eng. in Civil Engineering with a specialization in Structural Engineering from the University of Manitoba, and an Engineer-in-Training designation from APEGA.
- Thorough knowledge of engineering principles, structural design, and analysis of buildings, 3D modeling, building information modeling, technical report writing, and interpreting engineering drawings.
- Solid understanding of design codes such as NBCC Division B Part 4 and 9, CSA A23.3-14, CSA S16-19, CSSBI 58-2024, CSA-S6-06 (CHBDC), and CPCI Design Manual.

WORK EXPERIENCE

- **Structural EIT, Light-Weight Building Systems Inc.,** High River, AB *Feb. 2025- Present*
 - Perform structural design and develop detailed calculations for beams, columns, foundations, load-bearing steel stud wall panels, lintels, posts, and welded/bolted connections for residential and industrial projects, ensure compliance with relevant codes and industry standards, and prepare technical reports.
 - Conduct structural analysis, provide technical opinions to address design challenges, and utilize industry-standard software for analysis.
 - Conduct site visits and inspections at various stages of construction to evaluate conditions, coordinate with contractors and consultants, perform structural field reviews of drag struts before concrete pours by verifying rebar size, layout, and dimensions, and check slab elevations.
 - Supervise technologists and review shop drawings for load-bearing steel stud wall panels, beams, and columns.
- **Structural/Architectural Drafter, Splendid Homes Corp.,** Winnipeg, MB *May 2023- Oct. 2023*
 - Created architectural and structural packages based on provided designs, demolition plans, and detailed 3D BIM models for 10 single-family residential houses, each ranging from \$1 to \$5 million, using AutoCAD and Revit.
 - Conducted site visits to verify adherence to drawing sets and validated designs to meet structural specifications, ensuring compliance with the CSA O86 code.
 - Delivered Autodesk Revit training to junior designers.
- **Teaching Assistant, Bridge Engineering, University of Manitoba** *Jan. 2023- Apr. 2023*
 - Conducted weekly sessions for students to address their questions and resolve their problems.

TECHNICAL SKILLS

Modeling and Analysis

RISA 3D/Floor/Connection, Mathcad, WoodWorks,, SAP2000, Revit, AutoCAD, Robot Structural, Hilti PROFIS, Navisworks, MS Office (Excel, and Word), Bluebeam

VOLUNTEERING

- **Build Volunteer, Habitat for Humanity Manitoba** *Sep. 2024- Oct. 2024*
 - Assisted in carpentry tasks, installed drywall, and sheathed floors during the construction of 55 affordable homes for low-income families, ensuring compliance with safety standards under the guidance of construction leaders.

EDUCATION

University of Manitoba
Master of Engineering in Civil Engineering, GPA: 3.88/4.5
Recipient of International Graduate Student Entrance Scholarship (IGSES)

Sep. 2022- Oct. 2024

K. N. Toosi University of Technology
Master of Science in Construction Management, GPA: 3.88/4

Sep. 2019- Jun. 2022

Shahid Beheshti University
Bachelor of Science in Civil Engineering, GPA: 3.01/4

Sep. 2014- Feb. 2019

ACADEMIC PROJECTS

- **Development and Analysis of a Footbridge Using SAP2000** Sep. 2022- Dec. 2022
Finite Element Analysis coursework
 - Led a team of three to develop a detailed footbridge model in SAP2000, following plan and elevation drawings in compliance with CSA S6-06.
 - Analyzed the bridge model to determine the first 20 natural frequencies and their corresponding periods, as well as forces, moments, and maximum vertical displacement under various load conditions.
 - Documented the modeling approach for each structural element (beam, column, frame, slab, bearing, and brace).
 - Verified the accuracy and reliability of the work using detailed Excel-based calculations.
- **Design and 4D/5D BIM Simulation of a Residential Building ([Demo](#))** Feb. 2020-Jun. 2020
Building Information Modeling coursework
 - Designed the architectural/structural plan of a residential building based on LOD 300 in AutoCAD and Revit.
 - Detected and resolved clashes by overlaying structural and sprinkler plans using Navisworks and animated the construction process using the Navisworks animator module.
 - Executed 4D BIM simulation by developing a Work Breakdown Structure (WBS) in MS Project and 5D BIM simulation by conducting detailed quantity takeoff using the Navisworks quantification module.

PUBLICATIONS

The Use of 3D-Printed PA6-GF Components for the Construction of Structural Specimens
Reza Nazar Shahsavanai, Graziano Fiorillo Dec. 2024